PART 1 · MODULE A

AI fundamentals in plain language

The ideas behind every AI tool you will use, explained first in everyday words.

A1. Artificial intelligence

Simple explanation. A computer system that does tasks that normally need human judgement, such as reading text, recognising pictures or answering questions.

A2. Machine learning and deep learning

Machine learning: teaching a computer to find patterns from examples. Like a teacher who shows many photos of cats and dogs until a child can tell them apart. **Deep learning:** machine learning with many layers of simple calculations, used for images, speech and large language models.

A3. Generative AI and large language models (LLMs)

Simple explanation. Generative AI creates new text, images, audio or video. An LLM is a text model trained on large amounts of writing. It predicts the next piece of text, token by token (see the Development and IT textbook for the basics of an API).

A4. Prompt engineering

Writing clear instructions for a model: role, context, task, input, rules and the output format. A weak prompt says 'write a post'. A strong prompt says who the post is for, the length, the tone, the facts that must appear, and the format.

A5. AI agents and agentic AI

Simple explanation. An agent is an AI that can take steps with tools: read an email, look up a sheet, draft a reply and wait for approval. Agentic AI means systems that plan and act over several steps.

A6. Retrieval-augmented generation (RAG)

Simple explanation. Before the AI answers, the system finds the most relevant passages in your own documents and gives them to the model. The answer then cites its sources. Like an open-book exam.

A7. APIs and webhooks

An API lets your software ask an AI model a question and get a reply. A webhook is the reverse: a service calls your address when something happens (a form is submitted).

A8. Traditional automation versus AI automation

Traditional automation	AI automation
Follows fixed rules: if the amount is over 500, send to the manager.	Understands text and decides a category: is this email a complaint, an enquiry or spam?
Predictable and easy to test.	Flexible but can be wrong; needs validation and review.

A9. Hallucinations and model limits

A model can state false facts with confidence. It may not know recent events, your prices or your policies. Fix: give it the source text (RAG), ask it to say 'not found' when unsure, and check every number.

A10. Data privacy and responsible AI

- Know where data goes and whether it is used to train models on your plan.
- Remove personal data before sending text to a model when you can.
- Disclose AI-generated content where your market or platform requires it.
- Keep a human owner for decisions that affect people.

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